

Toqa Gamal Mohamed

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OBJECTIVE

Recent Computer & Systems Engineering graduate seeking a QA Engineer role, with hands-on experience in software testing and functional verification (UVM/SystemVerilog). Through the DEPI Software Testing Track, I learned **manual testing** across the **SDLC**, including functional, regression, integration, and exploratory testing, test case design, and the defect lifecycle, along with **API testing** using **Postman**, **ISTQB Foundation** topics, **Jira**, and **Selenium**. In a Jira-managed healthcare booking project, I designed **37 manual test cases** from user-story requirements using **Boundary Value Analysis** and **Equivalence Partitioning**. In my graduation project, I built **UVM testbenches** (SystemVerilog, QuestaSim) that uncovered RTL bugs and reached **95%+** functional and code coverage. Eager to grow from **manual testing** into **automation** and quality engineering in an **Agile/Scrum team**.

EDUCATION

Faculty of Engineering, Ain Shams University – Computer & Systems Engineering

Class of 2026

Cumulative GPA: Very Good (3.1/4.0)

PROFESSIONAL EXPERIENCE

Graduation Project – Digital Verification of PCIe 5.0 Transaction Layer

Sep 2025 – Jun 2026

Ain Shams University, Faculty of Engineering – Computer & Systems Engineering

Supervised by Prof. Dr. Ayman Wahba, Ain Shams University.

Summary of Project :

- Conducted a project study of the PCIe 5.0 specification to define the functional behavior of the Transaction Layer and produced a comprehensive block-level architecture capturing the protocol's four core functionalities.
- Built UVM-based testbenches (SystemVerilog, QuestaSim) for four PCIe 5.0 TL RX modules, developing scoreboards, reference models, drivers, monitors, and functional coverage models.
- Identified and documented RTL bugs through targeted verification scenarios, including boundary/off-by-one errors, incorrect CRC complement logic, missing enable gating, and incorrect address reconstruction.
- Achieved **95%+ functional and code coverage across all verified modules**, with results documented in a comprehensive coverage report.

My Role:

Within the 8-member team, I was responsible for the functional verification of four PCIe 5.0 Transaction Layer RX modules:

tl_rx_write_handler_ecrc, tl_rx_error_check_unsupported_req, tl_rx_error_check_unexpected_cpl, and tl_rx_error_check_receiver_overflow.

- Built the complete UVM verification environment for each module – including scoreboards, reference models, drivers, monitors, sequences, and functional coverage models – and validated the RTL implementation against the PCIe 5.0 specification.
- Designed test scenarios that uncovered and documented intentional RTL bugs (off-by-one boundary conditions, missing enable gating, incorrect BAR address reconstruction, and incorrect CRC complement logic), debugged compilation/simulation issues in QuestaSim, and iteratively improved functional and code coverage to 95%+ across all four modules.
- Analyzed the coverage reports and contributed to the technical write-up and presentation of the verification results.

Full Stack Web Development (PHP Track) – ITI

Jun /July 2025

Final Project: Online Restaurant Management System

- Built a full-stack restaurant ordering and reservation platform using PHP, MySQL, HTML, CSS, Bootstrap, and JavaScript.

- Designed a relational database schema (menu items, cart items, orders, order items, reservations, customers) and implemented core workflows: browsing the menu, adding items to cart, checkout, and order confirmation.
- Developed an admin dashboard for managing menu items, categories, users, and orders, including order status updates and reservation tracking.
- Implemented dynamic cart updates and form validation using JavaScript.

PROJECT

Healthcare Appointment Booking (Manual Testing) | Jira, Manual Testing | DEPI

Sep 2026

- Analyzed user story REQ-FUN-05 (Patient Details & Promo Code Engine) and wrote 37 test cases in Jira covering patient selection, beneficiary name/phone validation, and promo code rules (length, allowed characters, trim, uppercase normalization, invalid codes).
- Defined preconditions, steps, test data, and expected results for each test case, applying Boundary Value Analysis and Equivalence Partitioning to the field limits (3–50 chars, 11-digit phone, 15-char promo).
- Executed test cases and identified and reported defects in Jira, documenting steps to reproduce, expected vs. actual results, and severity.
- Took part in test case peer reviews within an Agile sprint (HL Sprint 1).

Testing (Movie Recommendation System) | Java, JUnit 5

Spring 2025

- Designed and implemented 350+ black-box test cases (JUnit 5) for a Java-based movie recommendation system, covering input validation, parsing, and recommendation logic.
- Applied Boundary Value Analysis, Equivalence Class Partitioning, Negative Testing, and Decision Table Testing to validate data formats and recommendation behavior against spec requirements.

MIPS 32-bit Processor | Verilog

Fall 2024

- Architected a structural 32-bit MIPS processor by connecting register files, control units, and memory blocks.
- Validated processor functionality with a custom-designed test bench.

Library Management System | MySQL

Spring 2025

- Mapped an EER diagram from ERD Plus into a relational schema and implemented the system using SQL queries by using phpMyAdmin

CPU Scheduler | C++, Qt6

Spring 2025

- Simulated FCFS, Round Robin, SJF, and Priority scheduling algorithms with real-time Gantt chart visualization using Qt.
- Achieved a dynamic and interactive scheduling interface to visualize process transitions and CPU time slices.

Online Shopping System | Java, JavaFX

Spring 2023

- Designed an e-commerce platform with object-oriented principles and exception handling.
- Enabled browsing and purchasing with a user-friendly GUI using JavaFX

Simple 2-bit ALU | ICs, 7-Segments, Decoders

Fall 2022

- Constructed a functional 2-bit ALU supporting add, subtract, AND, and OR operations

COURSES

- **Software Testing Track – Digital Egypt Pioneers Initiative (DEPI)** [Ongoing – Expected DEC 2026]
Undergoing comprehensive training in software testing throughout the SDLC, covering ISTQB Foundation Level topics: testing fundamentals, static testing techniques, test analysis and design, and managing test activities in Agile environments.
Gaining hands-on experience with test tools including Jira, Postman, and Selenium, and building foundational Java programming skills.
Track concludes with a Capstone Project applying end-to-end software testing methodologies
- **Introduction to Java – Coursera**
Explore fundamental Java programming concepts, including syntax, control structures, and object-oriented design.

SKILLS

Testing: Functional, Regression, Integration, Exploratory & Black-Box/White-Box Testing, Test Case Design (BVA, ECP, Decision Table), Bug Lifecycle & Defect Reporting, STLC/SDLC, Agile/Scrum

Tools: Jira, Postman (API Testing), Selenium, JUnit 5

Databases: SQL, MySQL, phpMyAdmin

Programming Languages: Java, Python, C, C++, MIPS Assembly

Verification: UVM, SystemVerilog, Verilog, QuestaSim, ModelSim, Functional/Code Coverage Analysis

Software Development: OOP, Data Structures and Algorithms

Developer Tools: Qt6, VS Code, NetBeans, IntelliJ, EDA playground